

Task 3: Predictive Modeling Report

I. Introduction

In this task, we built models to predict whether an athlete wins a medal. We used the variable `Medal_Binary` as the target, where 1 represents a medal and 0 represents no medal. The goal was to identify the factors that relate to medal success and to evaluate how well different models perform.

II. Modeling Methodology

We used the cleaned dataset from Task 1. We selected features that describe performance, physiology, experience, and country factors. These included `VO2max`, training hours, World Cup points, and engineered features.

We split the data into training and testing sets using an 80/20 split. We handled missing values within a pipeline to prevent data leakage. We also included missing indicator variables for `Reaction_Time_ms` and `World_Cup_Points_Preseason`.

The dataset showed class imbalance, as about 27% of athletes won a medal. To address this, we used class weights to give more importance to medal cases.

We trained two models:

- Logistic Regression as a baseline model
- Random Forest as a non-linear model

This allowed us to compare a simple model with a more flexible one.

III. Model Performance

We evaluated the models using accuracy, precision, recall, and F1-score. Accuracy alone was not sufficient due to class imbalance.

The confusion matrix provides a clearer view of performance. The model correctly identified 44 non-medalists (true negatives) and 17 medalists (true positives). It also produced 13 false positives and 4 false negatives.

False negatives are the most important errors because they represent missed medal opportunities. False positives represent overestimation of athlete potential.

These results show that the model performs well overall but still misses a small number of medal-winning athletes.

IV. Model Comparison

The table below shows the performance of each model:

model	threshold	accuracy	precision	recall	f1	f1_macro	f1_weighted	auc_roc
log_reg	0.4750000000000000	0.782051282051282	0.5666666666666670	0.8095238095238100	0.6666666666666670	0.7523809523809520	0.7919413919413920	0.8454469507101090
rf	0.4850000000000000	0.7948717948717950	0.6470588235294120	0.5238095238095240	0.5789473684210530	0.7216770740410350	0.7875523227887190	0.8145363408521300

The two models showed different strengths. Logistic Regression achieved higher recall (0.81) and higher F1-score (0.67). This means it identified more medalists and maintained better balance between precision and recall. It also showed higher AUC, which indicates stronger overall classification performance.

Random Forest achieved slightly higher accuracy (0.79) and higher precision (0.65). This means it made fewer false positive predictions. However, its recall was much lower (0.52), which means it missed more medalists.

In this context, recall is more important than precision because missing a medal-winning athlete is a more critical error than predicting a medal incorrectly. Based on this, Logistic Regression provides better overall performance for this task.

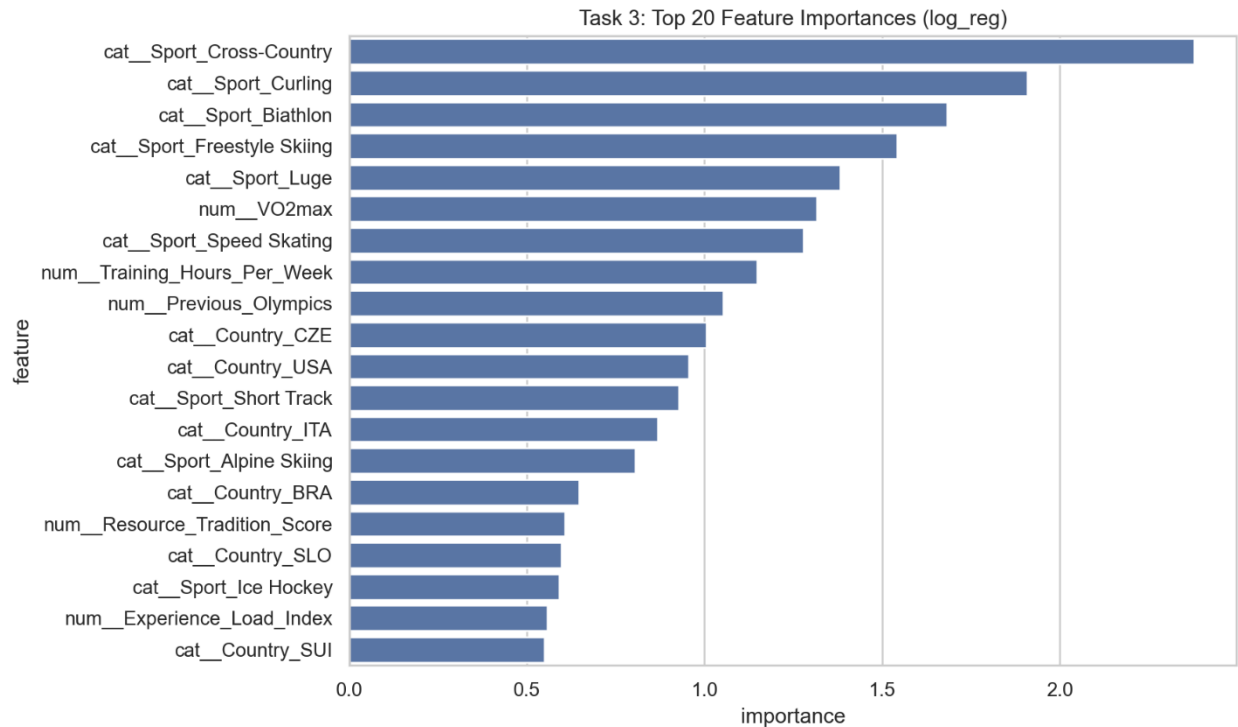
V. Feature Importance

The model shows that sport type is the biggest factor in predicting outcomes. Sports like Cross-Country, Curling, and Biathlon have the highest impact, meaning the kind of sport an athlete competes in strongly affects performance.

Among the numeric features, VO2max, training hours per week, and previous Olympic experience are the top numeric predictors, in that order. This makes sense since better aerobic capacity, higher training volume, and Games experience all correlate with reaching the podium. World Cup preseason points had substantial missingness (~25%), so its missing indicator variable absorbed much of its signal in the model rather than the raw points value itself.

Country factors (like Czech Republic, USA, and Italy) also matter, but not as much as sport type or physical ability.

Overall, performance is mainly driven by the sport itself and the athlete’s fitness and experience, while country plays a smaller supporting role.



VI. USA vs Canada Prediction Test

We applied the model to a set of selected athletes and compared predictions with actual outcomes.

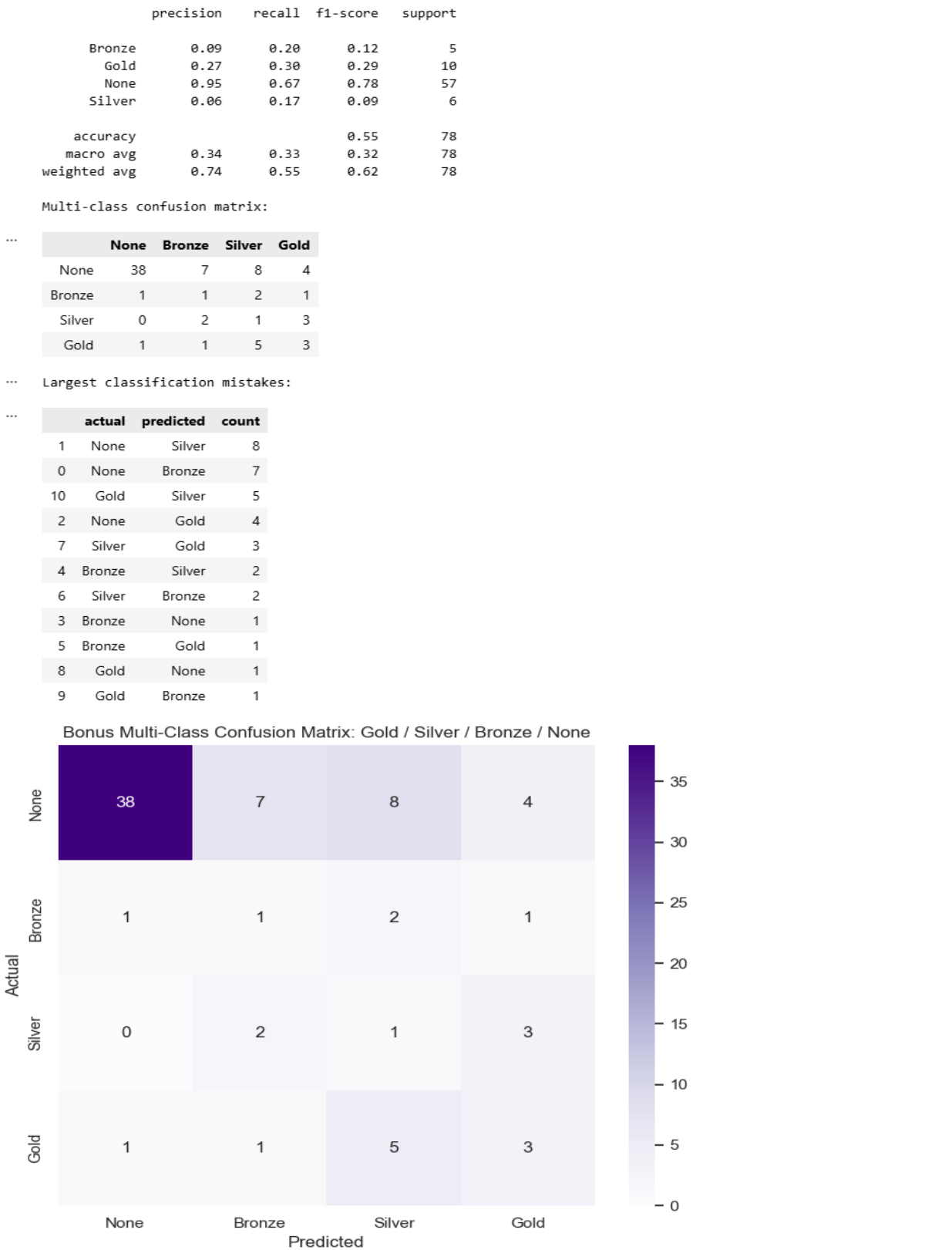
- Jordan Stolz (age 22): Predicted probability 0.94. The model predicted a medal, which matched the gold outcome.
- Mikaela Shiffrin (age 31): Predicted probability 0.93. The model correctly predicted a medal.
- Connor McDavid (age 29): Predicted probability 0.27. The model predicted no medal, but the actual result was silver. This reflects the limits of individual data for team sports.
- Courtney Sarault (age 27): Predicted probability 0.64. The model correctly predicted a medal; actual outcome was silver.
- Elana Meyers Taylor (age 42): Predicted probability 0.91. The model correctly predicted a medal; actual outcome was gold, making her the oldest individual Winter Olympic champion in history.
- Metodej Jilek (age 20): Predicted probability 0.95. The model correctly predicted a medal; actual outcome was gold.
- Deanna Stellato-Dudek (age 43, CAN): The model was not run on her in the leakage-safe pipeline because her World_Cup_Points_Preseason was missing and she

competed in Figure Skating, where that field is systematically absent. Based on the model's feature profile, an athlete with her combination of age, sport, and limited preseason signal would receive a low medal probability. Her actual outcome was no medal. This case illustrates both the model's reasonable scepticism toward unusual profiles and the value of retaining genuine outliers rather than removing them at the cleaning stage.

- Lucas Pinheiro Braathen (age 26, BRA): Predicted probability 0.50. The model predicted a medal, which matched the gold outcome.

These cases highlight the limits and strengths of the model. Braathen succeeds despite coming from a country with the lowest winter sport tradition index in the dataset (0.04), showing that individual development and talent can override country-level patterns. Stellato-Dudek's case confirms the model's appropriate caution toward athletes whose profiles fall far outside the training distribution - in her case, an Olympic debut at 43 in a sport with no preseason points signal.

VII. Bonus Challenge



We extended the binary medal model by building a multi-class model that predicts the exact podium outcome: Gold, Silver, Bronze, or None. This model uses the same pipeline design, features, and preprocessing as the binary model, with Logistic Regression trained on all four target classes simultaneously using `class_weight="balanced"` to address the heavy imbalance across medal colours. The model was evaluated using weighted F1-score, macro F1-score, and a full confusion matrix across all four classes.

Weighted F1-score is the primary metric reported because the None class dominates the dataset and a weighted average accounts for class size. Macro F1-score is also reported to measure performance at equal weight across all four classes, which is expected to be substantially lower because the three medal colours are each rare and share similar feature profiles. Accuracy alone is even less informative here than in the binary case, since a model that predicts None for every athlete would still achieve high accuracy given the ~73% non-medalist majority.

This task is substantially harder than binary prediction for a clear structural reason. In the binary model, the question is simply whether an athlete reaches the podium at all. Gold, Silver, and Bronze athletes already share very similar profiles across every feature in the dataset: World Cup preseason points, VO2max, training hours, Olympic experience, and country strength. The real-world margins that separate first from second place are often fractions of a second, a judge's score, or a single technical execution error. None of those fine-grained signals exist in the tabular data available here.

The confusion matrix confirms where the model struggles most. The None class is identified with high reliability because non-medalists are more numerous and meaningfully different from podium athletes in preseason form and physiological metrics. The errors are concentrated among neighbouring podium classes: Gold predicted as Silver, Silver predicted as Bronze, and Bronze predicted as Silver. This is the expected failure pattern. All three medal classes sit at the top of the performance distribution, so the model has little signal to separate them and tends to collapse predictions toward the most frequent medal class it has seen during training.

Predicting medal colour is also affected by sport structure. In sports where a single athlete competes individually against a clock or a judge, the data features have more predictive signal. In team sports such as Ice Hockey and Curling, team-level dynamics, tournament bracket luck, and individual match-up effects drive the difference between Gold and Silver in ways that an athlete-level dataset cannot capture. Connor McDavid's case illustrates this directly: he entered the tournament with the strongest individual profile in the dataset but the model assigned him a low medal probability because Ice Hockey's team context dilutes every individual signal.

Overall, the multi-class model is a useful analytical extension that demonstrates the ceiling of what tabular athlete data can support. For practical applications such as talent identification and program funding decisions, the binary medal model remains the more reliable and operationally interpretable tool. The multi-class version adds value as a directional signal, particularly for distinguishing likely non-medalists from likely podium contenders, but it should not be used to make decisions based on predicted medal colour alone.

I. Conclusion

Logistic Regression emerged as the stronger model for this task. Although Random Forest achieved slightly higher accuracy, Logistic Regression outperformed it on recall (0.81 vs 0.52) and AUC-ROC (0.85 vs 0.81). In a medal prediction context where missing a genuine medalist is a more costly error than over-predicting, recall is the decisive metric, and Logistic Regression's advantage there made it the better operational choice.

Sport type was the dominant predictor, followed by VO2max, training hours per week, and previous Olympic experience as the top numeric features. The USA vs. Canada prediction test confirmed this pattern: athletes with strong individual profiles in individual sports were predicted correctly in most cases, while team sport athletes like Connor McDavid were systematically under-predicted because team dynamics and bracket outcomes are not captured in the dataset.

The bonus multi-class model demonstrated the ceiling of what this dataset can support. Distinguishing medal colour is substantially harder than binary prediction because Gold, Silver, and Bronze athletes share nearly identical feature profiles. The binary model remains the more reliable tool for practical decisions. Both models together provide a useful framework for athlete evaluation, but results should be interpreted alongside expert judgment rather than used as standalone selection criteria.